

[GETNET]ORACLE DATABASE 21C INSTALLATION ON REDHAT 8 (OL8)

A instalação do Oracle Database 21c, neste host, se dará depois de remover a instalação anterior, em 19c. Sendo assim, vamos fazer alguns passos de remoção antes da instalação.

1- REMOÇÃO DA INSTALAÇÃO PRÉVIA

```
[root@GNCASNHL08985 u01]# cd /tmp
[root@GNCASNHL08985 tmp]# rm -rf *
[root@GNCASNHL08985 tmp]# rm -rf /etc/oraInst.loc
[root@GNCASNHL08985 tmp]# cd /etc/oracle
[root@GNCASNHL08985 oracle]# rm -rf *
[root@GNCASNHL08985 oracle]# rm -rf /etc/oratab
[root@GNCASNHL08985 oracle]# rm -rf /tmp/.oracle

[root@GNCASNHL08985 oracle]# /usr/lib/methods/ucfgacfsctl -l ofscctl
-bash: /usr/lib/methods/ucfgacfsctl: No such file or directory
[root@GNCASNHL08985 oracle]# /usr/lib/methods/ucfgadmctl -l advmctl
-bash: /usr/lib/methods/ucfgadmctl: No such file or directory
[root@GNCASNHL08985 oracle]# /usr/lib/methods/udefacfsctl -l ofscctl
-bash: /usr/lib/methods/udefacfsctl: No such file or directory
[root@GNCASNHL08985 oracle]# /usr/lib/methods/udefadmctl -l advmctl
-bash: /usr/lib/methods/udefadmctl: No such file or directory
[root@GNCASNHL08985 oracle]# /usr/sbin/rmauth -h oracle
-bash: /usr/sbin/rmauth: No such file or directory
[root@GNCASNHL08985 oracle]# rmrole oracle_devmgmt
-bash: rmrole: command not found
[root@GNCASNHL08985 oracle]# setkst
-bash: setkst: command not found
[root@GNCASNHL08985 oracle]# rm /usr/lib/drivers/oracle*
rm: cannot remove '/usr/lib/drivers/oracle*': No such file or directory
[root@GNCASNHL08985 oracle]# rm /usr/lib/methods/*advm* /usr/lib/methods/*acfs*
rm: cannot remove '/usr/lib/methods/*advm*': No such file or directory
rm: cannot remove '/usr/lib/methods/*acfs*': No such file or directory
[root@GNCASNHL08985 oracle]# rm -rf /sbin/helpers/acfs
[root@GNCASNHL08985 oracle]# rm /usr/sbin/acfsutil* /usr/sbin/advmutil*
rm: cannot remove '/usr/sbin/acfsutil*': No such file or directory
rm: cannot remove '/usr/sbin/advmutil*': No such file or directory
[root@GNCASNHL08985 oracle]# rm /sbin/acfsutil* /sbin/advmutil*
rm: cannot remove '/sbin/acfsutil*': No such file or directory
rm: cannot remove '/sbin/advmutil*': No such file or directory

[root@GNCASNHL08985 oracle]# cd /u01/
[root@GNCASNHL08985 u01]# ls -la
total 4
drwxrwxr-x  3 grid oinstall 17 Jan 30 12:51 .
dr-xr-xr-x. 23 root root   4096 Jan 30 15:33 ..
drwxr-xr-x  6 grid oinstall 65 Jan 30 12:51 app
[root@GNCASNHL08985 u01]# rm -rf app/

[root@GNCASNHL08985 u01]# oracleasm listdisks
ARCH01_PSUPLAY
ARCH02_PSUPLAY
ARCH03_PSUPLAY
ARCH04_PSUPLAY
DATA01_AUX_PSUPLAY
DATA01_PSUPLAY
DATA02_AUX_PSUPLAY
DATA02_PSUPLAY
DATA03_AUX_PSUPLAY
DATA03_PSUPLAY
DATA04_AUX_PSUPLAY
DATA04_PSUPLAY

[root@GNCASNHL08985 u01]# oracleasm deletedisk ARCH01_PSUPLAY
```

```

Clearing disk header: done
Dropping disk: done
[root@GNCASNHL08985 u01]# oracleasm deletedisk ARCH02_PSUPLAY
Clearing disk header: done
Dropping disk: done
[root@GNCASNHL08985 u01]# oracleasm deletedisk ARCH03_PSUPLAY
Clearing disk header: done
Dropping disk: done
[root@GNCASNHL08985 u01]# oracleasm deletedisk ARCH04_PSUPLAY
Clearing disk header: done
Dropping disk: done
[root@GNCASNHL08985 u01]# oracleasm deletedisk DATA01_AUX_PSUPLAY
Clearing disk header: done
Dropping disk: done
[root@GNCASNHL08985 u01]# oracleasm deletedisk DATA01_PSUPLAY
Clearing disk header: done
Dropping disk: done
[root@GNCASNHL08985 u01]# oracleasm deletedisk DATA02_AUX_PSUPLAY
Clearing disk header: done
Dropping disk: done
[root@GNCASNHL08985 u01]# oracleasm deletedisk DATA02_PSUPLAY
Clearing disk header: done
Dropping disk: done
[root@GNCASNHL08985 u01]# oracleasm deletedisk DATA03_AUX_PSUPLAY
Clearing disk header: done
Dropping disk: done
[root@GNCASNHL08985 u01]# oracleasm deletedisk DATA03_PSUPLAY
Clearing disk header: done
Dropping disk: done
[root@GNCASNHL08985 u01]# oracleasm deletedisk DATA04_AUX_PSUPLAY
Clearing disk header: done
Dropping disk: done

[root@GNCASNHL08985 u01]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...

[root@GNCASNHL08985 u01]# oracleasm listdisks

[root@GNCASNHL08985 u01]# fdisk -l
Disk /dev/sda: 37 GiB, 39728447488 bytes, 77594624 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 95993A3B-301E-46D2-9653-14795FF1B965

Device      Start    End  Sectors  Size Type
/dev/sda1    2048 1230847 1228800 600M EFI System
/dev/sda2 1230848 3327999 2097152 1G Linux filesystem
/dev/sda3 3328000 77594590 74266591 35.4G Linux filesystem

Disk /dev/sdc: 100 GiB, 107374182400 bytes, 209715200 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xc270015d

Device  Boot Start    End  Sectors  Size Id Type
/dev/sdc1    2048 209715199 209713152 100G 83 Linux

Disk /dev/sdb: 162 GiB, 173946175488 bytes, 339738624 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos

```

Disk identifier: 0x607a5588

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdb1		2048	339738623	339736576	162G	83	Linux

Disk /dev/sdd: 145 GiB, 155692564480 bytes, 304087040 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x98aba68c

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdd1		2048	76021759	76019712	36.3G	83	Linux
/dev/sdd2		76021760	152043519	76021760	36.3G	83	Linux
/dev/sdd3		152043520	228065279	76021760	36.3G	83	Linux
/dev/sdd4		228065280	304087039	76021760	36.3G	83	Linux

Disk /dev/sde: 145 GiB, 155692564480 bytes, 304087040 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x55766659

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sde1		2048	76021759	76019712	36.3G	83	Linux
/dev/sde2		76021760	152043519	76021760	36.3G	83	Linux
/dev/sde3		152043520	228065279	76021760	36.3G	83	Linux
/dev/sde4		228065280	304087039	76021760	36.3G	83	Linux

Disk /dev/sdf: 37 GiB, 39728447488 bytes, 77594624 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa731e2be

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdf1		2048	19398655	19396608	9.3G	83	Linux
/dev/sdf2		19398656	38797311	19398656	9.3G	83	Linux
/dev/sdf3		38797312	58195967	19398656	9.3G	83	Linux
/dev/sdf4		58195968	77594623	19398656	9.3G	83	Linux

Disk /dev/mapper/rhel-root: 16.4 GiB, 17620271104 bytes, 34414592 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/rhel-swap: 18 GiB, 19327352832 bytes, 37748736 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/vg_ggs-lv_ggs: 100 GiB, 107369988096 bytes, 209707008 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/vg_oracle-lv_u01: 80 GiB, 85899345920 bytes, 167772160 sectors
Units: sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/vg_oracle-lv_u02: 80 GiB, 85899345920 bytes, 167772160 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
[root@GNCASNHL08985 u01]#

```
[root@GNCASNHL08985 u01]# dd if=/dev/zero of=/dev/sdd1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0653455 s, 784 kB/s
[root@GNCASNHL08985 u01]# dd if=/dev/zero of=/dev/sdd2 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0365656 s, 1.4 MB/s
[root@GNCASNHL08985 u01]# dd if=/dev/zero of=/dev/sdd3 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0222565 s, 2.3 MB/s
[root@GNCASNHL08985 u01]# dd if=/dev/zero of=/dev/sdd4 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0452459 s, 1.1 MB/s
[root@GNCASNHL08985 u01]#
[root@GNCASNHL08985 u01]# dd if=/dev/zero of=/dev/sde1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0437933 s, 1.2 MB/s
[root@GNCASNHL08985 u01]# dd if=/dev/zero of=/dev/sde2 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0492956 s, 1.0 MB/s
[root@GNCASNHL08985 u01]# dd if=/dev/zero of=/dev/sde3 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00955686 s, 5.4 MB/s
[root@GNCASNHL08985 u01]# dd if=/dev/zero of=/dev/sde4 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0150364 s, 3.4 MB/s
[root@GNCASNHL08985 u01]#
[root@GNCASNHL08985 u01]# dd if=/dev/zero of=/dev/sdf1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.038898 s, 1.3 MB/s
[root@GNCASNHL08985 u01]# dd if=/dev/zero of=/dev/sdf2 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0275645 s, 1.9 MB/s
[root@GNCASNHL08985 u01]# dd if=/dev/zero of=/dev/sdf3 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0345229 s, 1.5 MB/s
[root@GNCASNHL08985 u01]# dd if=/dev/zero of=/dev/sdf4 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0186243 s, 2.7 MB/s
```

2- ARQUIVOS UTILIZADOS NA INSTALAÇÃO

Para instalação do Oracle 21c, baixamos para o repositório de mídias os seguintes arquivos:

```
wget https://repository.getnet.com.br/binarios/oracle/database/LINUX.X64_213000_db_home.zip      - Instalador Grid (21.3)
wget https://repository.getnet.com.br/binarios/oracle/database/LINUX.X64_213000_grid_home.zip   - Instalador Oracle (21.3)
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p6880880_190000_Linux-x86-64.zip - Opatch (v41)
```

3- CRIAÇÃO DOS DISCOS DE ASM

Com a limpeza feita no host, agora, propriamente podemos pegar os primeiros passos de instalação:

NOTA: nenhuma diferença para o 19c

```
[root@GNCASNHL08985 u01]# oracleasm createdisk DATA01_PSUPLAY /dev/sdd1
Writing disk header: done
Instantiating disk: done
[root@GNCASNHL08985 u01]# oracleasm createdisk DATA02_PSUPLAY /dev/sdd2
Writing disk header: done
Instantiating disk: done
[root@GNCASNHL08985 u01]# oracleasm createdisk DATA03_PSUPLAY /dev/sdd3
Writing disk header: done
Instantiating disk: done
[root@GNCASNHL08985 u01]# oracleasm createdisk DATA04_PSUPLAY /dev/sdd4
Writing disk header: done
Instantiating disk: done
[root@GNCASNHL08985 u01]#
[root@GNCASNHL08985 u01]# oracleasm createdisk ARCH01_PSUPLAY /dev/sde1
Writing disk header: done
Instantiating disk: done
[root@GNCASNHL08985 u01]# oracleasm createdisk ARCH02_PSUPLAY /dev/sde2
Writing disk header: done
Instantiating disk: done
[root@GNCASNHL08985 u01]# oracleasm createdisk ARCH03_PSUPLAY /dev/sde3
Writing disk header: done
Instantiating disk: done
[root@GNCASNHL08985 u01]# oracleasm createdisk ARCH04_PSUPLAY /dev/sde4
Writing disk header: done
Instantiating disk: done
[root@GNCASNHL08985 u01]#
[root@GNCASNHL08985 u01]# oracleasm createdisk DATA01_AUX_PSUPLAY /dev/sdf1
Writing disk header: done
Instantiating disk: done
[root@GNCASNHL08985 u01]# oracleasm createdisk DATA02_AUX_PSUPLAY /dev/sdf2
Writing disk header: done
Instantiating disk: done
[root@GNCASNHL08985 u01]# oracleasm createdisk DATA03_AUX_PSUPLAY /dev/sdf3
Writing disk header: done
Instantiating disk: done
[root@GNCASNHL08985 u01]# oracleasm createdisk DATA04_AUX_PSUPLAY /dev/sdf4
Writing disk header: done
Instantiating disk: done

[root@GNCASNHL08985 u01]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...

[root@GNCASNHL08985 u01]# oracleasm listdisks
ARCH01_PSUPLAY
ARCH02_PSUPLAY
ARCH03_PSUPLAY
ARCH04_PSUPLAY
DATA01_AUX_PSUPLAY
DATA01_PSUPLAY
DATA02_AUX_PSUPLAY
DATA02_PSUPLAY
DATA03_AUX_PSUPLAY
DATA03_PSUPLAY
DATA04_AUX_PSUPLAY
DATA04_PSUPLAY
```

4- INSTALAÇÃO DOS PACOTES PARA XDISPLAY E AJUSTES DO USER ADM

Não são necessários para instalação via ansible, mas, como fiz a instalação manual, para registro, seguimos:

```
[root@GNCASNHL08985 u01]# yum install xorg-x11-xauth
Updating Subscription Management repositories.
MongoDB 5 EL7                               37 kB/s | 1.5 kB    00:00
Red Hat Satellite Maintenance 6.13 for RHEL 8 x86_64 (RPMs) 66 kB/s | 4.2 kB    00:00
Zabbix - Rhel6 I386                          34 kB/s | 1.5 kB    00:00
Red Hat Satellite Capsule 6.13 for RHEL 8 x86_64 (RPMs) 94 kB/s | 4.5 kB    00:00
Grafana                                     33 kB/s | 1.5 kB    00:00
Zabbix - Rhel6 x86_64                       55 kB/s | 1.5 kB    00:00
Zabbix 4.0                                  21 kB/s | 1.5 kB    00:00
Zabbix - Rhel8                             42 kB/s | 1.5 kB    00:00
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs) 117 kB/s | 4.5 kB    00:00
Red Hat Satellite Client 6 for RHEL 8 x86_64 (RPMs) 76 kB/s | 3.8 kB    00:00
Red Hat Satellite 6.13 for RHEL 8 x86_64 (RPMs) 46 kB/s | 4.5 kB    00:00
Red Hat Satellite Utils 6.13 for RHEL 8 x86_64 (RPMs) 74 kB/s | 4.2 kB    00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs) 115 kB/s | 4.1 kB    00:00
Zabbix 5.0 frontend                         34 kB/s | 1.5 kB    00:00
Zabbix - Rhel7                             50 kB/s | 1.5 kB    00:00
Zabbix 6.0 RHEL7                           30 kB/s | 1.5 kB    00:00
Dependencies resolved.
```

```
=====
Package           Architecture Version      Repository                               Size
=====
Installing:
xorg-x11-xauth    x86_64      1:1.0.9-12.el8 rhel-8-for-x86_64-appstream-rpms      39 k
=====
```

Transaction Summary

```
=====
Install 1 Package
```

Total download size: 39 k

Installed size: 63 k

Is this ok [y/N]: y

Downloading Packages:

```
xorg-x11-xauth-1.0.9-12.el8.x86_64.rpm      319 kB/s | 39 kB    00:00
-----
```

```
Total                                     315 kB/s | 39 kB    00:00
```

Running transaction check

Transaction check succeeded.

Running transaction test

Transaction test succeeded.

Running transaction

```
Preparing      :                               1/1
Installing      : xorg-x11-xauth-1:1.0.9-12.el8.x86_64      1/1
Running scriptlet: xorg-x11-xauth-1:1.0.9-12.el8.x86_64      1/1
Verifying       : xorg-x11-xauth-1:1.0.9-12.el8.x86_64      1/1
```

Installed products updated.

Installed:

```
xorg-x11-xauth-1:1.0.9-12.el8.x86_64
```

Complete!

```
[root@GNCASNHL08985 u01]# yum install xterm
```

Updating Subscription Management repositories.

```
MongoDB 5 EL7                               24 kB/s | 1.5 kB    00:00
Red Hat Satellite Maintenance 6.13 for RHEL 8 x86_64 (RPMs) 94 kB/s | 4.2 kB    00:00
Zabbix - Rhel6 I386                          21 kB/s | 1.5 kB    00:00
Red Hat Satellite Capsule 6.13 for RHEL 8 x86_64 (RPMs) 80 kB/s | 4.5 kB    00:00
Grafana                                     39 kB/s | 1.5 kB    00:00
Zabbix - Rhel6 x86_64                       38 kB/s | 1.5 kB    00:00
Zabbix 4.0                                  18 kB/s | 1.5 kB    00:00
Zabbix - Rhel8                             43 kB/s | 1.5 kB    00:00
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs) 128 kB/s | 4.5 kB    00:00
Red Hat Satellite Client 6 for RHEL 8 x86_64 (RPMs) 83 kB/s | 3.8 kB    00:00
Red Hat Satellite 6.13 for RHEL 8 x86_64 (RPMs) 130 kB/s | 4.5 kB    00:00
Red Hat Satellite Utils 6.13 for RHEL 8 x86_64 (RPMs) 48 kB/s | 4.2 kB    00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs) 75 kB/s | 4.1 kB    00:00
Zabbix 5.0 frontend                         21 kB/s | 1.5 kB    00:00
Zabbix - Rhel7                             61 kB/s | 1.5 kB    00:00
Zabbix 6.0 RHEL7                           26 kB/s | 1.5 kB    00:00
Dependencies resolved.
```

```
=====
Package           Architecture Version      Repository                               Size
=====
Installing:
xterm              x86_64       331-2.el8    rhel-8-for-x86_64-appstream-rpms        529 k
Installing dependencies:
xterm-resize       x86_64       331-2.el8    rhel-8-for-x86_64-appstream-rpms        39 k
Installing weak dependencies:
xorg-x11-fonts-misc noarch       7.5-19.el8   rhel-8-for-x86_64-appstream-rpms        5.8 M

Transaction Summary
=====
Install 3 Packages

Total download size: 6.3 M
Installed size: 8.3 M
Is this ok [y/N]: y
Downloading Packages:
(1/3): xterm-resize-331-2.el8.x86_64.rpm          537 kB/s | 39 kB   00:00
(2/3): xterm-331-2.el8.x86_64.rpm                4.2 MB/s | 529 kB  00:00
(3/3): xorg-x11-fonts-misc-7.5-19.el8.noarch.rpm  20 MB/s | 5.8 MB   00:00
-----
Total                                           22 MB/s | 6.3 MB   00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing      :                               1/1
Installing    : xterm-resize-331-2.el8.x86_64          1/3
Installing    : xorg-x11-fonts-misc-7.5-19.el8.noarch  2/3
Running scriptlet: xorg-x11-fonts-misc-7.5-19.el8.noarch 2/3
Installing    : xterm-331-2.el8.x86_64                 3/3
Running scriptlet: xterm-331-2.el8.x86_64                 3/3
Verifying     : xorg-x11-fonts-misc-7.5-19.el8.noarch  1/3
Verifying     : xterm-331-2.el8.x86_64                 2/3
Verifying     : xterm-resize-331-2.el8.x86_64          3/3
Installed products updated.

Installed:
xorg-x11-fonts-misc-7.5-19.el8.noarch    xterm-331-2.el8.x86_64    xterm-resize-331-2.el8.x86_64

Complete!
[root@GNCASNHL08985 u01]#

[root@GNCASNHL08985 home]# mkdir adm.as58172
[root@GNCASNHL08985 home]# chown -R adm.as58172 adm.as58172
[root@GNCASNHL08985 home]# chmod -R 775 adm.as58172
```

5- CRIAÇÃO DE PROFILES E DIRETÓRIOS

Até este ponto, tudo igual:

```
[root@GNCASNHL08985 u01]# cat /home/oracle/db_profile
export ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=/u01/app/oracle/product/21.3/dbhome_1
export ORACLE_SID=PSUPLAY
export PATH=/u01/app/oracle/product/21.3/dbhome_1/bin:$PATH
export NLS_LANG="AMERICAN_AMERICA.WE8MSWIN1252"
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:/usr/lib:/ggs
export OGGHOME=/ggs

[root@GNCASNHL08985 u01]# cat /home/grid/grid_profile
export ORACLE_BASE=/u01/app/grid
export ORACLE_HOME=/u01/app/21.3/grid
export ORACLE_SID=+ASM
export PATH=/u01/app/21.3/grid/bin:$PATH

[root@GNCASNHL08985 u01]# mkdir -p /u01/app/oracle
[root@GNCASNHL08985 u01]# mkdir -p /u01/app/oracle/product/21.3/dbhome_1
[root@GNCASNHL08985 u01]# mkdir -p /u01/app/grid
[root@GNCASNHL08985 u01]# mkdir -p /u01/app/21.3/grid
[root@GNCASNHL08985 u01]# chown oracle:oinstall /u01/app/oracle
[root@GNCASNHL08985 u01]# chown oracle:oinstall /u01/app/oracle/product/21.3/dbhome_1
[root@GNCASNHL08985 u01]# chown grid:oinstall /u01/app/grid
[root@GNCASNHL08985 u01]# chown grid:oinstall /u01/app/21.3/grid
[root@GNCASNHL08985 u01]# chmod -R 775 /u01
[root@GNCASNHL08985 u01]# mkdir -p /u01/app/oralInventory
[root@GNCASNHL08985 u01]# chown grid:oinstall /u01/app/oralInventory
[root@GNCASNHL08985 u01]# chmod 775 /u01/app/oralInventory
```

5- AJUSTES PRÉ INSTALL

O ajuste do /etc/hosts sempre foi necessário, por vezes encontramos hosts sem o apontamento do próprio IP neste arquivo e pegamos erros na hora de aplicar PSU. Válido lembrar que tem que ser o mesmo nome do /etc/hostname.

```
[root@GNCASNHL08985 home]# cat /etc/hostname
GNCASNHL08985

[root@GNCASNHL08985 home]# cat /etc/hosts
127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4
::1 localhost loopback
180.135.58.23 GNCASNHL08985 GNCASNHL08985.getnet.local
22.240.224.245 gncasswn05738 gncasswn05738.wt.getnet.local
```

De acordo com os pacotes necessários, detectei apenas 1 que fez upgrade:

```
[root@GNCASNHL08985 home]# dnf install -y glibc
Updating Subscription Management repositories.
MongoDB 5 EL7                                45 kB/s | 1.5 kB    00:00
Red Hat Satellite Maintenance 6.13 for RHEL 8 x86_64 (RPMs) 58 kB/s | 4.2 kB    00:00
Zabbix - Rhel6 I386                          46 kB/s | 1.5 kB    00:00
Red Hat Satellite Capsule 6.13 for RHEL 8 x86_64 (RPMs) 74 kB/s | 4.5 kB    00:00
Grafana                                       56 kB/s | 1.5 kB    00:00
Zabbix - Rhel6 x86_64                       43 kB/s | 1.5 kB    00:00
Zabbix 4.0                                  31 kB/s | 1.5 kB    00:00
Zabbix - Rhel8                              55 kB/s | 1.5 kB    00:00
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs) 72 kB/s | 4.5 kB    00:00
Red Hat Satellite Client 6 for RHEL 8 x86_64 (RPMs) 78 kB/s | 3.8 kB    00:00
Red Hat Satellite 6.13 for RHEL 8 x86_64 (RPMs) 118 kB/s | 4.5 kB    00:00
Red Hat Satellite Utils 6.13 for RHEL 8 x86_64 (RPMs) 103 kB/s | 4.2 kB    00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs) 106 kB/s | 4.1 kB    00:00
Zabbix 5.0 frontend                        56 kB/s | 1.5 kB    00:00
Zabbix - Rhel7                              51 kB/s | 1.5 kB    00:00
Zabbix 6.0 RHEL7                          53 kB/s | 1.5 kB    00:00
Package glibc-2.28-236.el8.7.x86_64 is already installed.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Upgrading:				
glibc	x86_64	2.28-236.el8_9.12	rhel-8-for-x86_64-baseos-rpms	2.2 M
glibc-common	x86_64	2.28-236.el8_9.12	rhel-8-for-x86_64-baseos-rpms	1.0 M

```
glibc-devel          x86_64      2.28-236.el8_9.12  rhel-8-for-x86_64-baseos-rpms  86 k
glibc-gconv-extra    x86_64      2.28-236.el8_9.12  rhel-8-for-x86_64-baseos-rpms  1.6 M
glibc-headers        x86_64      2.28-236.el8_9.12  rhel-8-for-x86_64-baseos-rpms  491 k
glibc-langpack-en    x86_64      2.28-236.el8_9.12  rhel-8-for-x86_64-baseos-rpms  829 k
libnsl               x86_64      2.28-236.el8_9.12  rhel-8-for-x86_64-baseos-rpms  108 k

Transaction Summary
=====
Upgrade 7 Packages

Total download size: 6.2 M
Downloading Packages:
(1/7): glibc-devel-2.28-236.el8_9.12.x86_64.rpm          1.2 MB/s | 86 kB   00:00
(2/7): glibc-common-2.28-236.el8_9.12.x86_64.rpm         8.8 MB/s | 1.0 MB  00:00
(3/7): glibc-gconv-extra-2.28-236.el8_9.12.x86_64.rpm    17 MB/s | 1.6 MB  00:00
(4/7): glibc-2.28-236.el8_9.12.x86_64.rpm               12 MB/s | 2.2 MB   00:00
(5/7): glibc-headers-2.28-236.el8_9.12.x86_64.rpm       6.5 MB/s | 491 kB  00:00
(6/7): libnsl-2.28-236.el8_9.12.x86_64.rpm              2.5 MB/s | 108 kB  00:00
(7/7): glibc-langpack-en-2.28-236.el8_9.12.x86_64.rpm    12 MB/s | 829 kB   00:00
-----
Total                                     27 MB/s | 6.2 MB   00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Upgrading      : glibc-common-2.28-236.el8_9.12.x86_64          1/14
  Upgrading      : glibc-gconv-extra-2.28-236.el8_9.12.x86_64     2/14
  Running scriptlet: glibc-gconv-extra-2.28-236.el8_9.12.x86_64   2/14
  Upgrading      : glibc-langpack-en-2.28-236.el8_9.12.x86_64     3/14
  Running scriptlet: glibc-2.28-236.el8_9.12.x86_64              4/14
  Upgrading      : glibc-2.28-236.el8_9.12.x86_64                4/14
  Running scriptlet: glibc-2.28-236.el8_9.12.x86_64              4/14
  Running scriptlet: glibc-headers-2.28-236.el8_9.12.x86_64      5/14
  Upgrading      : glibc-headers-2.28-236.el8_9.12.x86_64        5/14
  Upgrading      : glibc-devel-2.28-236.el8_9.12.x86_64          6/14
  Running scriptlet: glibc-devel-2.28-236.el8_9.12.x86_64        6/14
  Upgrading      : libnsl-2.28-236.el8_9.12.x86_64               7/14
  Running scriptlet: glibc-devel-2.28-236.el8.7.x86_64           8/14
  Cleanup        : glibc-devel-2.28-236.el8.7.x86_64            8/14
  Cleanup        : libnsl-2.28-236.el8.7.x86_64                 9/14
  Cleanup        : glibc-headers-2.28-236.el8.7.x86_64          10/14
  Cleanup        : glibc-2.28-236.el8.7.x86_64                 11/14
  Cleanup        : glibc-langpack-en-2.28-236.el8.7.x86_64      12/14
  Cleanup        : glibc-gconv-extra-2.28-236.el8.7.x86_64      13/14
  Running scriptlet: glibc-gconv-extra-2.28-236.el8.7.x86_64    13/14
  Cleanup        : glibc-common-2.28-236.el8.7.x86_64           14/14
  Running scriptlet: glibc-common-2.28-236.el8.7.x86_64         14/14
  Running scriptlet: glibc-common-2.28-236.el8_9.12.x86_64      14/14
  Verifying      : glibc-2.28-236.el8_9.12.x86_64              1/14
  Verifying      : glibc-2.28-236.el8.7.x86_64                 2/14
  Verifying      : glibc-common-2.28-236.el8_9.12.x86_64       3/14
  Verifying      : glibc-common-2.28-236.el8.7.x86_64          4/14
  Verifying      : glibc-devel-2.28-236.el8_9.12.x86_64        5/14
  Verifying      : glibc-devel-2.28-236.el8.7.x86_64           6/14
  Verifying      : glibc-gconv-extra-2.28-236.el8_9.12.x86_64  7/14
  Verifying      : glibc-gconv-extra-2.28-236.el8.7.x86_64     8/14
  Verifying      : glibc-headers-2.28-236.el8_9.12.x86_64     9/14
  Verifying      : glibc-headers-2.28-236.el8.7.x86_64        10/14
  Verifying      : glibc-langpack-en-2.28-236.el8_9.12.x86_64  11/14
  Verifying      : glibc-langpack-en-2.28-236.el8.7.x86_64     12/14
  Verifying      : libnsl-2.28-236.el8_9.12.x86_64             13/14
  Verifying      : libnsl-2.28-236.el8.7.x86_64                14/14
Installed products updated.

Upgraded:
  glibc-2.28-236.el8_9.12.x86_64 glibc-common-2.28-236.el8_9.12.x86_64 glibc-devel-2.28-236.el8_9.12.x86_64 glibc-gconv-extra-2.28-236.el8_9.12.x86_64 glibc-headers-2.28-236.el8_9.12.x86_64 glibc-langpack-en-2.28-236.el8_9.12.x86_64 libnsl-2.28-236.el8_9.12.x86_64
```

Complete!

Quanto aos ajustes listados abaixo, todos permanecem iguais.

- /etc/sysctl.conf
- group
- user
- SELINUX
- firewall

6- INSTALL DO GRID INFRASTRUCTURE

Até o presente momento, instalação igual ao 19c, não vou tentar instalar nenhum pacote, apenas validar se o instalador pedirá algo mais assim, saberemos que podemos seguir com os pacotes do template 19c.

```
[grid@GNCASNHL08985 grid]$ cp /u02/oracle/patches/LINUX.X64_213000_grid_home.zip .
[grid@GNCASNHL08985 grid]$ unzip LINUX.X64_213000_grid_home.zip
```

```
...
inflating: rhp/patch/rhp_root_owned_files.txt
inflating: rhp/patch/rootrhp.pl
linking: bin/lbuilder -> ../nls/lbuilder/lbuilder
linking: lib/libocci.so -> libocci.so.21.1
linking: lib/libagtsh.so -> libagtsh.so.1.0
linking: lib/libjavavm.a -> ../javavm/jdk/jdk8/lib/libjavavm.a
linking: lib/libclntsh.so -> libclntsh.so.21.1
linking: javavm/lib/jce.jar -> ../javavm/jdk/jdk8/lib/jce.jar
linking: jdk/bin/ControlPanel -> jcontrol
linking: javavm/admin/cbp.jar -> ../javavm/jdk/jdk8/admin/cbp.jar
linking: lib/libclntshcore.so -> libclntshcore.so.21.1
linking: jdk/jre/bin/ControlPanel -> jcontrol
linking: javavm/admin/libjtcjt.so -> ../javavm/jdk/jdk8/admin/libjtcjt.so
linking: javavm/admin/classes.bin -> ../javavm/jdk/jdk8/admin/classes.bin
linking: javavm/lib/security/cacerts -> ../javavm/jdk/jdk8/lib/security/cacerts
linking: javavm/lib/security/README.txt -> ../javavm/jdk/jdk8/lib/security/README.txt
linking: javavm/lib/sunjce_provider.jar -> ../javavm/jdk/jdk8/lib/sunjce_provider.jar
linking: javavm/lib/security/java.security -> ../javavm/jdk/jdk8/lib/security/java.security
linking: jdk/jre/lib/amd64/server/libjsig.so -> ../libjsig.so
```

```
extracting: install/.img.bin
```

```
inflating: install/files.lst
```

```
finishing deferred symbolic links:
```

```
bin/lbuilder -> ../nls/lbuilder/lbuilder
lib/libocci.so -> libocci.so.21.1
lib/libagtsh.so -> libagtsh.so.1.0
lib/libjavavm.a -> ../javavm/jdk/jdk8/lib/libjavavm.a
lib/libclntsh.so -> libclntsh.so.21.1
javavm/lib/jce.jar -> ../javavm/jdk/jdk8/lib/jce.jar
jdk/bin/ControlPanel -> jcontrol
javavm/admin/cbp.jar -> ../javavm/jdk/jdk8/admin/cbp.jar
lib/libclntshcore.so -> libclntshcore.so.21.1
jdk/jre/bin/ControlPanel -> jcontrol
javavm/admin/libjtcjt.so -> ../javavm/jdk/jdk8/admin/libjtcjt.so
javavm/admin/classes.bin -> ../javavm/jdk/jdk8/admin/classes.bin
javavm/lib/security/cacerts -> ../javavm/jdk/jdk8/lib/security/cacerts
javavm/lib/security/README.txt -> ../javavm/jdk/jdk8/lib/security/README.txt
javavm/lib/sunjce_provider.jar -> ../javavm/jdk/jdk8/lib/sunjce_provider.jar
javavm/lib/security/java.security -> ../javavm/jdk/jdk8/lib/security/java.security
jdk/jre/lib/amd64/server/libjsig.so -> ../libjsig.so
```

```
[grid@GNCASNHL08985 grid]$
```

```
[grid@GNCASNHL08985 grid]$ hostnamectl
```

```
Static hostname: GNCASNHL08985
```

```
Icon name: computer-vm
```

```
Chassis: vm
```

```
Machine ID: baf42505e9f94dfb93bd9a35f1aeb73
```

```
Boot ID: 6f883d8e03ca456482eb045c3a14254d
```

```
Virtualization: vmware
```

```
Operating System: Red Hat Enterprise Linux 8.9 (Ootpa)
```

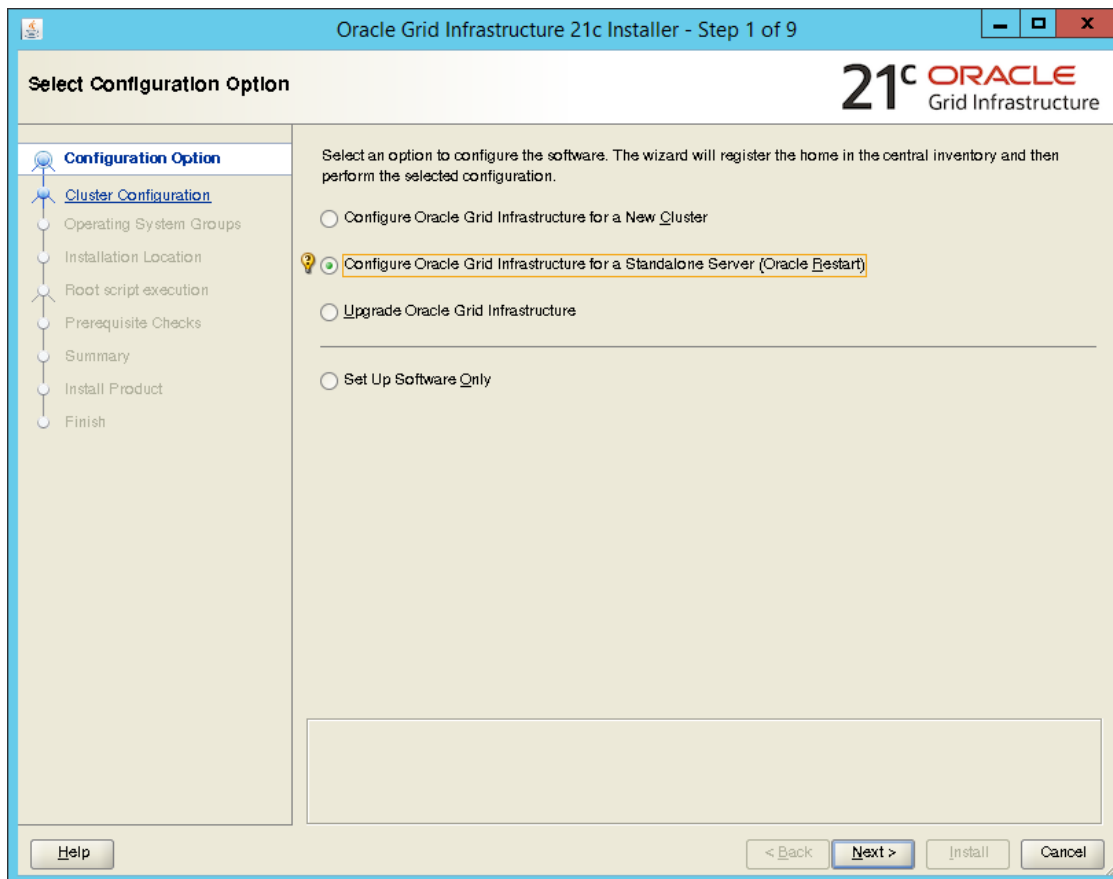
```
CPE OS Name: cpe:/o:redhat:enterprise_linux:8::baseos
```

```
Kernel: Linux 4.18.0-513.18.1.el8_9.x86_64
```

```
Architecture: x86-64
```

NOTA: como o S.O. já é muito novo, usar o parametro do CV_ASSUME_DISTID conforme abaixo:

```
[root@GNCASNHL08985 ~]# su - grid
Last login: Wed Mar 20 16:14:40 -03 2024 on pts/2
[grid@GNCASNHL08985 ~]$ . grid_profile
[grid@GNCASNHL08985 ~]$ export CV_ASSUME_DISTID=OL8
[grid@GNCASNHL08985 ~]$ xauth add GNCASNHL08985/unix:10 MIT-MAGIC-COOKIE-1 554a7b51cb67a1c9ceee6635812cdd72
xauth: file /home/grid/.Xauthority does not exist
[grid@GNCASNHL08985 ~]$ xauth add GNCASNHL08985/unix:10 MIT-MAGIC-COOKIE-1 554a7b51cb67a1c9ceee6635812cdd72
[grid@GNCASNHL08985 ~]$ export DISPLAY=localhost:10.0
[grid@GNCASNHL08985 ~]$ cd $ORACLE_HOME
[grid@GNCASNHL08985 grid]$ pwd
/u01/app/21.3/grid
[grid@GNCASNHL08985 ~]$ gridSetup.sh
```



Lembrar de indicar:

O Nome correto do Diskgroup DATA, redundancy "External" e alterar o "Change Discovey Path"

Oracle Grid Infrastructure 21c Installer - Step 4 of 11

21c ORACLE
Grid Infrastructure

Specify Management Options

Configuration Option

Create ASM Disk Group

ASM Password

Management Options

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

You can configure to have this instance of Oracle Grid Infrastructure and Oracle Automatic Storage Management to be managed by Enterprise Manager Cloud Control. Specify the details of the Cloud Control configuration to perform the registration.

☐ Register with Enterprise Manager (EM) Cloud Control

OMS host:

OMS port:

EM Admin User Name:

EM Admin Password:

Help

< Back

Next >

Install

Cancel

Oracle Grid Infrastructure 21c Installer - Step 5 of 11

Privileged Operating System Groups

Configuration Option

Create ASM Disk Group

ASM Password

Management Options

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Select the name of the operating system group, that you want to use for operating system authentication to Oracle Automatic Storage Management.

Oracle ASM Administrator (OSASM) Group

asmadmin

Oracle ASM DBA (OSDBA for ASM) Group

asmdba

Oracle ASM Operator (OSOPER for ASM) Group (Optional)

Help

< Back

Next >

Install

Cancel

Validar os diretórios do ORACLE_BASE e ORACLE_HOME de acordo com o profile criado:

Oracle Grid Infrastructure 21c Installer - Step 6 of 11

Specify Installation Location

Configuration Option

Create ASM Disk Group

ASM Password

Management Options

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Specify the Oracle base. The Oracle base directory for the Oracle Grid Infrastructure installation is the location where diagnostic and administrative logs, and other logs associated with Oracle ASM and Oracle Clusterware are stored. This location would also contain files pertaining to the configuration of Oracle Clusterware.

Oracle base:

/u01/app/grid

Browse...

This software directory is the Oracle Grid Infrastructure home directory.

Software location: /u01/app/21.3/grid

Help

< Back

Next >

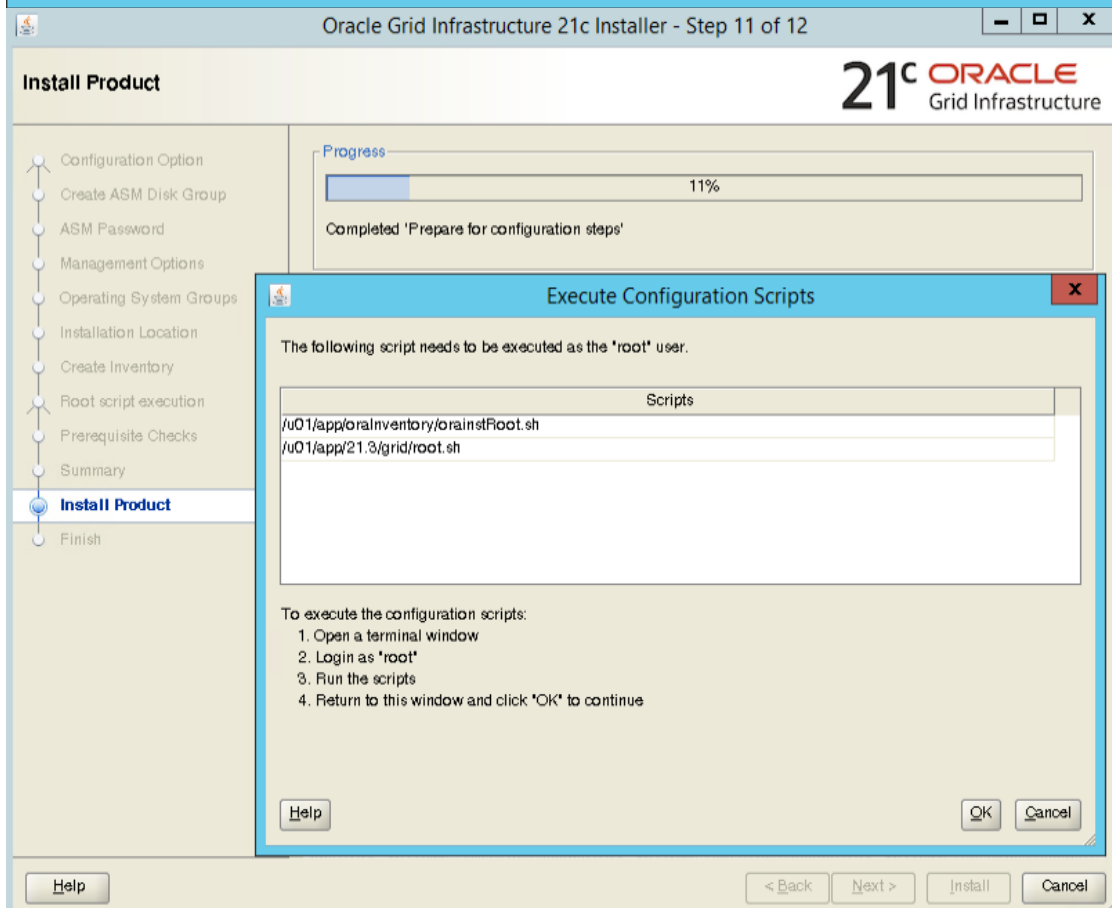
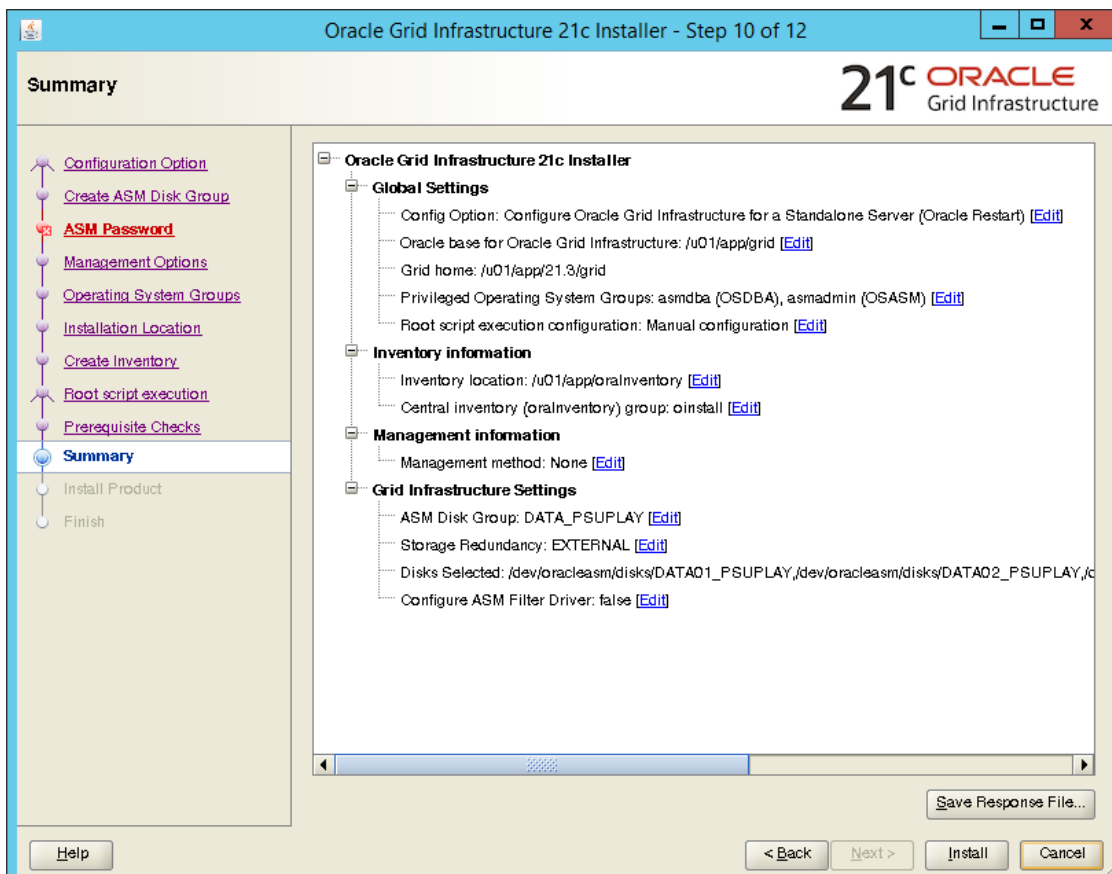
Install

Cancel

OracleInventory sempre padrão:

Core Assets/Documentation/GETNET -
DBA
ORAEX Cloud Consulting | GETNET | Feb 01, 2026 - 02:38am

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Scripts de root:

```
[root@GNCASNHL08985 home]# /u01/app/orainventory/orainstRoot.sh
Changing permissions of /u01/app/orainventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.

Changing groupname of /u01/app/orainventory to oinstall.
```

```
I he execution of the script is complete.

[root@GNCASNHL08985 home]# /u01/app/21.3/grid/root.sh
Performing root user operation.

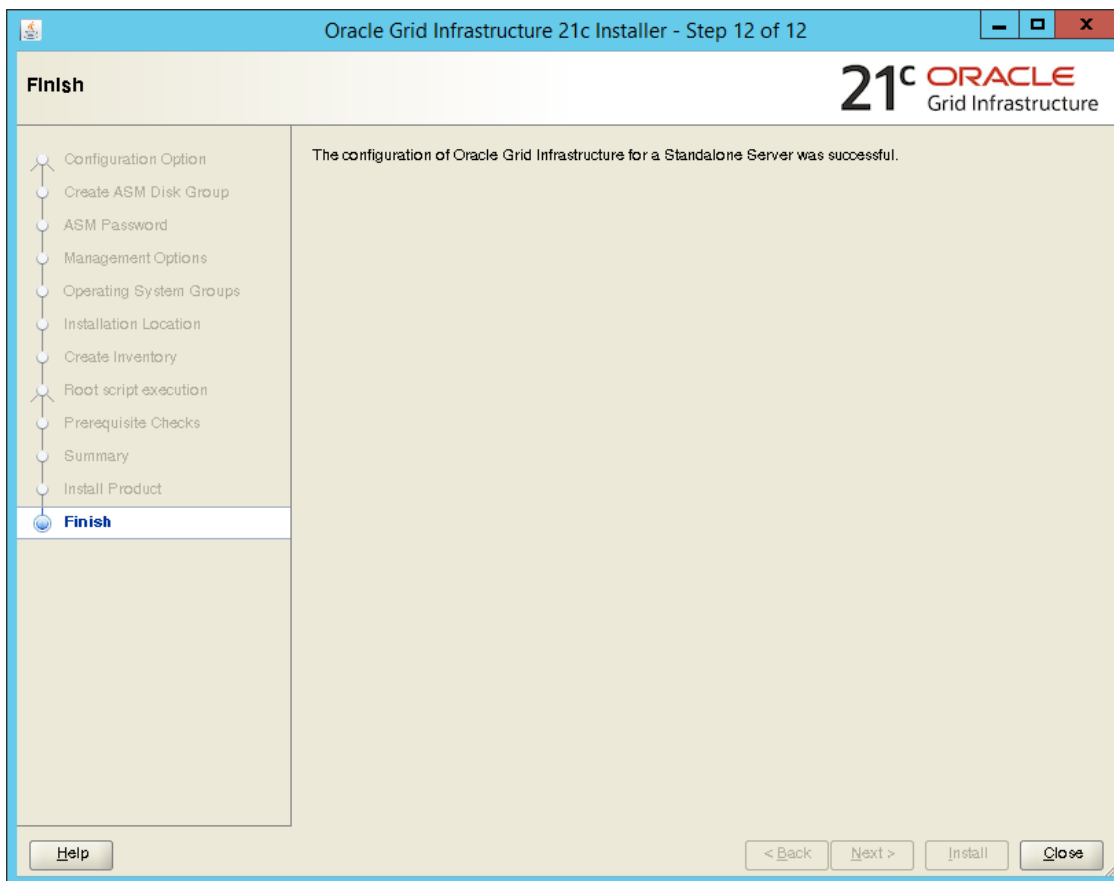
The following environment variables are set as:
  ORACLE_OWNER= grid
  ORACLE_HOME= /u01/app/21.3/grid

Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The file "oraenv" already exists in /usr/local/bin. Overwrite it? (y/n)
[n]: y
  Copying oraenv to /usr/local/bin ...
The file "coraenv" already exists in /usr/local/bin. Overwrite it? (y/n)
[n]: y
  Copying coraenv to /usr/local/bin ...

Creating /etc/oratab file...
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Using configuration parameter file: /u01/app/21.3/grid/crs/install/crsconfig_params
2024-03-20 16:59:23: Got permissions of file /u01/app/grid/crsdata/gncasnhl08985/crsconfig: 0775
2024-03-20 16:59:23: Got permissions of file /u01/app/grid/crsdata: 0775
2024-03-20 16:59:23: Got permissions of file /u01/app/grid/crsdata/gncasnhl08985: 0775
The log of current session can be found at:
  /u01/app/grid/crsdata/gncasnhl08985/crsconfig/roothas_2024-03-20_04-59-23PM.log
LOCAL ADD MODE
Creating OCR keys for user 'grid', privgrp 'oinstall'..
Operation successful.
LOCAL ONLY MODE
Successfully accumulated necessary OCR keys.
Creating OCR keys for user 'root', privgrp 'root'..
Operation successful.
CRS-4664: Node gncasnhl08985 successfully pinned.
2024/03/20 16:59:35 CLSRSC-330: Adding Clusterware entries to file 'oracle-ohasd.service'

gncasnhl08985 2024/03/20 17:00:19 /u01/app/grid/crsdata/gncasnhl08985/olr/backup_20240320_170019.olr 0
2024/03/20 17:00:19 CLSRSC-327: Successfully configured Oracle Restart for a standalone server

[root@GNCASNHL08985 home]# . /home/grid/grid_profile
[root@GNCASNHL08985 home]# crsctl status res -t
-----
Name          Target State      Server          State details
-----
Local Resources
-----
ora.ons
      OFFLINE OFFLINE    gncasnhl08985    STABLE
-----
Cluster Resources
-----
ora.cssd
  1    OFFLINE OFFLINE                STABLE
ora.diskmon
  1    OFFLINE OFFLINE                STABLE
ora.evmd
  1    ONLINE  ONLINE    gncasnhl08985    STABLE
-----
[root@GNCASNHL08985 home]#
```



Check ao fim do install:

```
[grid@GNCASNHL08985 grid]$ crsctl status res -t
```

Name	Target	State	Server	State details	
Local Resources					
ora.DATA_P\$UPLAY.dg					
	ONLINE	ONLINE	gncasnhl08985	STABLE	
ora.LISTENER.lsnr	ONLINE	ONLINE	gncasnhl08985	STABLE	
ora.asm	ONLINE	ONLINE	gncasnhl08985	Started,STABLE	
ora.ons	OFFLINE	OFFLINE	gncasnhl08985	STABLE	
Cluster Resources					
ora.cssd					
1	ONLINE	ONLINE	gncasnhl08985	STABLE	
ora.diskmon	1	OFFLINE	OFFLINE	STABLE	
ora.evmd	1	ONLINE	ONLINE	gncasnhl08985	STABLE

7- INSTALL DO ORACLE DATABASE

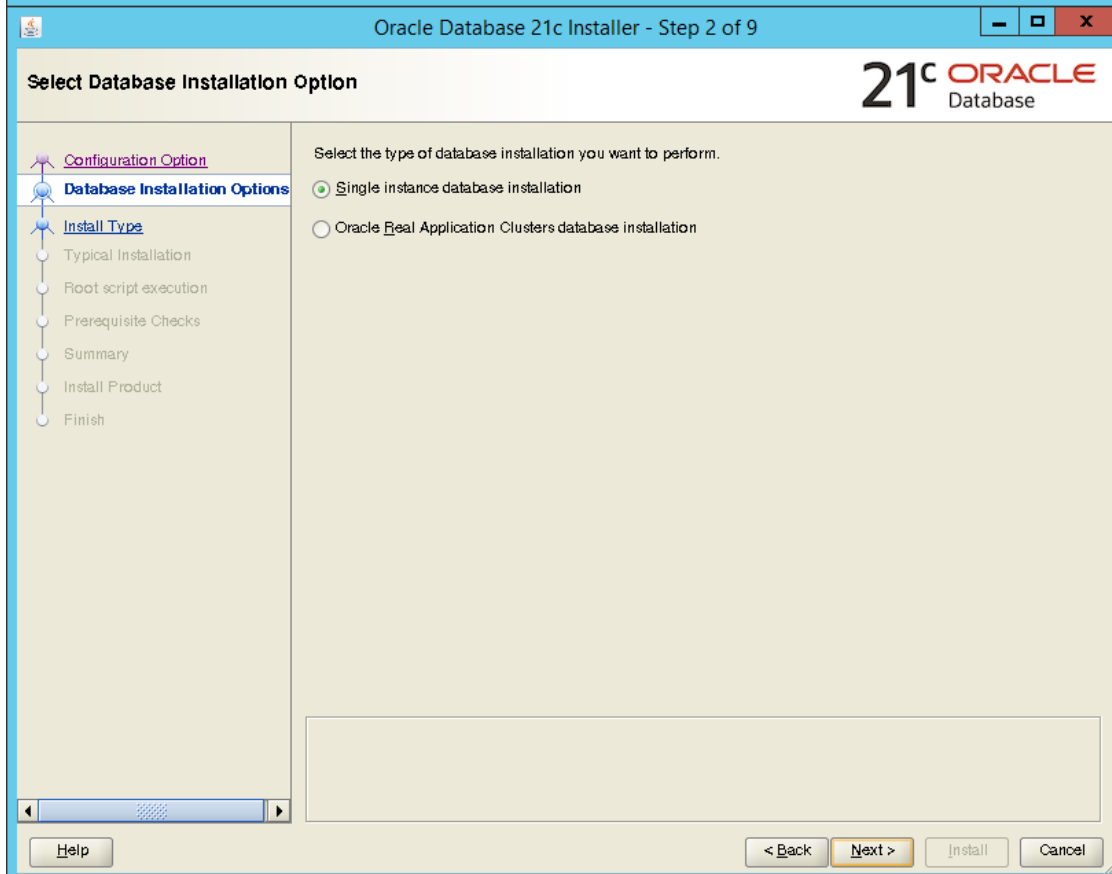
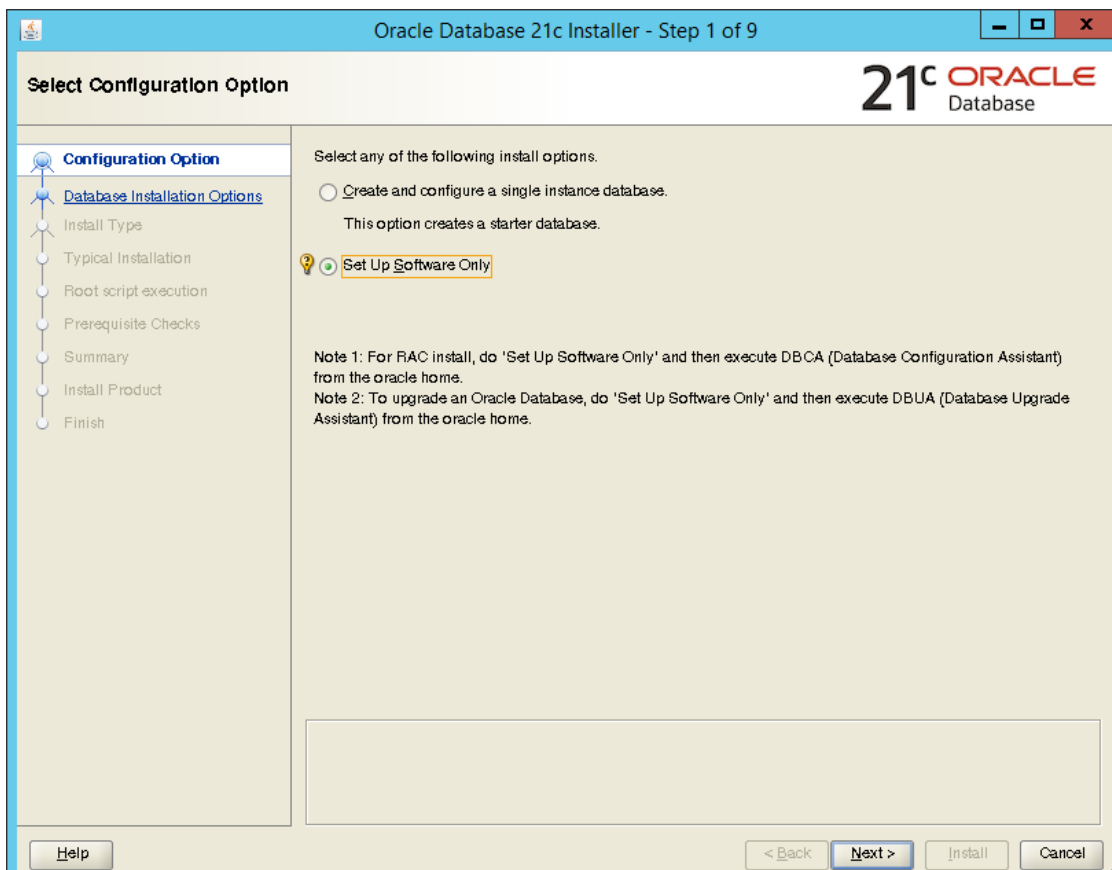
Exatamente igual ao 19c:

```

[oracle@GNCASNHL08985 dbhome_1]$ cp /u02/oracle/patches/LINUX.X64_213000_db_home.zip .
[oracle@GNCASNHL08985 dbhome_1]$ descompactar LINUX.X64_213000_db_home.zip
...
inflando: lib/libexpat.la
inflando: schagent.conf
vinculação: bin/lbuilder -> ../nls/lbuilder/lbuilder
vinculando: lib/libocci.so -> libocci.so.21.1
vinculando: lib/libjavavm.a -> ../javavm/jdk/jdk8/lib/libjavavm.a
vinculando: lib/libagtsh.so -> libagtsh.so.1.0
vinculando: lib/libclntsh.so -> libclntsh.so.21.1
vinculação: javavm/lib/jce.jar -> ../javavm/jdk/jdk8/lib/jce.jar
vinculação: javavm/admin/cbp.jar -> ../javavm/jdk/jdk8/admin/cbp.jar
vinculação: lib/libclntshcore.so -> libclntshcore.so.21.1
vinculação: jdk/bin/ControlPanel -> jcontrol
vinculação: jdk/jre/bin/ControlPanel -> jcontrol
vinculação: javavm/admin/libjtcjt.so -> ../javavm/jdk/jdk8/admin/libjtcjt.so
vinculação: javavm/admin/classes.bin -> ../javavm/jdk/jdk8/admin/classes.bin
vinculação: javavm/lib/security/cacerts -> ../javavm/jdk/jdk8/lib/security/cacerts
vinculando: javavm/lib/security/README.txt -> ../javavm/jdk/jdk8/lib/security/README.txt
vinculação: javavm/lib/sunjce_provider.jar -> ../javavm/jdk/jdk8/lib/sunjce_provider.jar
vinculando: javavm/lib/security/java.security -> ../javavm/jdk/jdk8/lib/security/java.security
vinculação: jdk/jre/lib/amd64/server/libjsig.so -> ../libjsig.so
extraindo: install/.img.bin
inflando: install/files.lst
finalizando links simbólicos diferenciados:
bin/lbuilder -> ../nls/lbuilder/lbuilder
lib/libocci.so -> libocci.so.21.1
lib/libjavavm.a -> ../javavm/jdk/jdk8/lib/libjavavm.a
lib/libagtsh.so -> libagtsh.so.1.0
lib/libclntsh.so -> libclntsh.so.21.1
javavm/lib/jce.jar -> ../javavm/jdk/jdk8/lib/jce.jar
javavm/admin/cbp.jar -> ../javavm/jdk/jdk8/admin/cbp.jar
lib/libclntshcore.so -> libclntshcore.so.21.1
jdk/bin/Painel de Controle -> jcontrol
jdk/jre/bin/Painel de Controle -> jcontrol
javavm/admin/libjtcjt.so -> ../javavm/jdk/jdk8/admin/libjtcjt.so
javavm/admin/classes.bin -> ../javavm/jdk/jdk8/admin/classes.bin
javavm/lib/security/cacerts -> ../javavm/jdk/jdk8/lib/security/cacerts
javavm/lib/security/README.txt -> ../javavm/jdk/jdk8/lib/security/README.txt
javavm/lib/sunjce_provider.jar -> ../javavm/jdk/jdk8/lib/sunjce_provider.jar
javavm/lib/security/java.security -> ../javavm/jdk/jdk8/lib/security/java.security
jdk/jre/lib/amd64/server/libjsig.so -> ../libjsig.so

[oracle@GNCASNHL08985 dbhome_1]$ xauth adicionar GNCASNHL08985/unix:10 MIT-MAGIC-COOKIE-1 554a7b51cb67a1c9ceee6635812cdd
72
[oracle@GNCASNHL08985 dbhome_1]$ enviar DISPLAY=localhost:10.0
[oracle@GNCASNHL08985 dbhome_1]$ cd $ORACLE_HOME
[oracle@GNCASNHL08985 dbhome_1]$ enviar CV_ASSUME_DISTID=OL8
[oracle@GNCASNHL08985 dbhome_1]$ ./runInstaller

```



Oracle Database 21c Installer - Step 3 of 10

21c ORACLE Database

Select Database Edition

Configuration Option

Database Installation Options

Database Edition

Installation Location

Operating System Groups

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Which database edition do you want to install?

☒ Enterprise Edition

Oracle Database 21c Enterprise Edition is a self-managing database that has the scalability, performance, high availability, and security features required to run the most demanding, mission-critical applications.

☐ Standard Edition 2

Oracle Database 21c Standard Edition 2 is a full-featured data management solution ideally suited to the needs of medium-sized businesses.

Help

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Next >

Install

Cancel

Oracle Database 21c Installer - Step 5 of 10

21c ORACLE Database

Privileged Operating System groups

Configuration Option

Database Installation Options

Database Edition

Installation Location

Operating System Groups

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

SYS privileges are required to create a database using operating system (OS) authentication. Membership in OS Groups grants the corresponding SYS privilege, eg. membership in OSDBA grants the SYSDBA privilege.

Database Administrator (OSDBA) group:

Database Operator (OSOPER) group (Optional):

Database Backup and Recovery (OSBACKUPDBA) group:

Data Guard administrative (OSDGDBA) group:

Encryption Key Management administrative (OSKMDBA) group:

Real Application Cluster administrative (OSRACDBA) group:

Help

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Install

Cancel

Oracle Database 21c Installer - Step 6 of 10

21c ORACLE Database

Root script execution configuration

Configuration Option

Database Installation Options

Database Edition

Installation Location

Operating System Groups

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

During the software configuration, certain operations have to be performed as "root" user. You can choose to have the installer perform these operations automatically by specifying inputs for one of the options below. The input specified will also be used by the installer to perform additional prerequisite checks.

☐ Automatically run configuration scripts

☒ Use "root" user credential

Password :

☐ Use sudo

Program path :

User name :

Password :

Help

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Install

Cancel

Oracle Database 21c Installer - Step 8 of 10

21c ORACLE Database

Summary

Configuration Option

Database Installation Options

Database Edition

Installation Location

Operating System Groups

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Oracle Database 21c Installer

Global settings

Database edition: Enterprise Edition (Set Up Software Only) [\[Edit\]](#)

Oracle base: /u01/app/oracle [\[Edit\]](#)

Software location: /u01/app/oracle/product/21.3/dbhome_1

Privileged Operating System groups: dba (OSDBA), oper (OSOPER), dba (OSBACKUPDBA), dba (OS

Root script execution configuration: Manual configuration [\[Edit\]](#)

Help

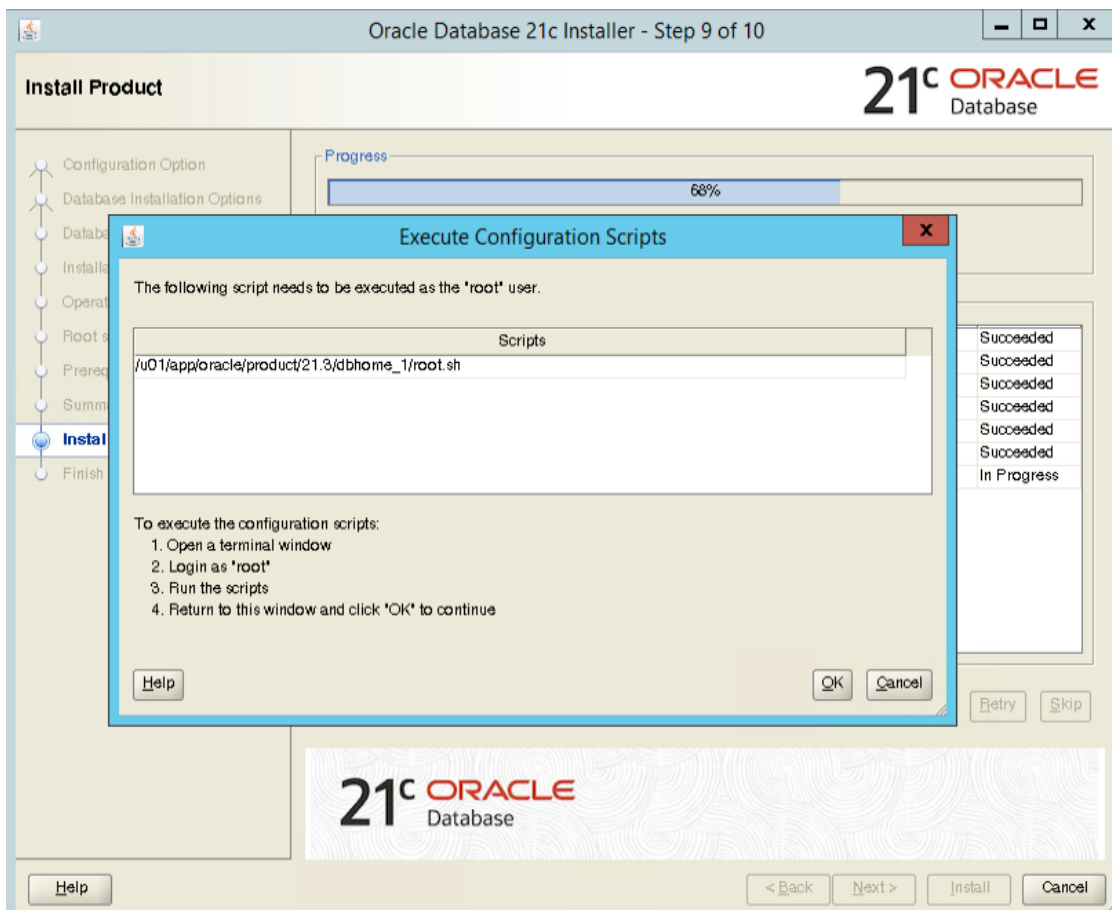
< Back

Next >

Install

Cancel

Save Response File...



Script de root:

```
[root@GNCASNHL08985 home]# /u01/app/oracle/product/21.3/dbhome_1/root.sh
Performing root user operation.
```

The following environment variables are set as:

```
ORACLE_OWNER= oracle
ORACLE_HOME= /u01/app/oracle/product/21.3/dbhome_1
```

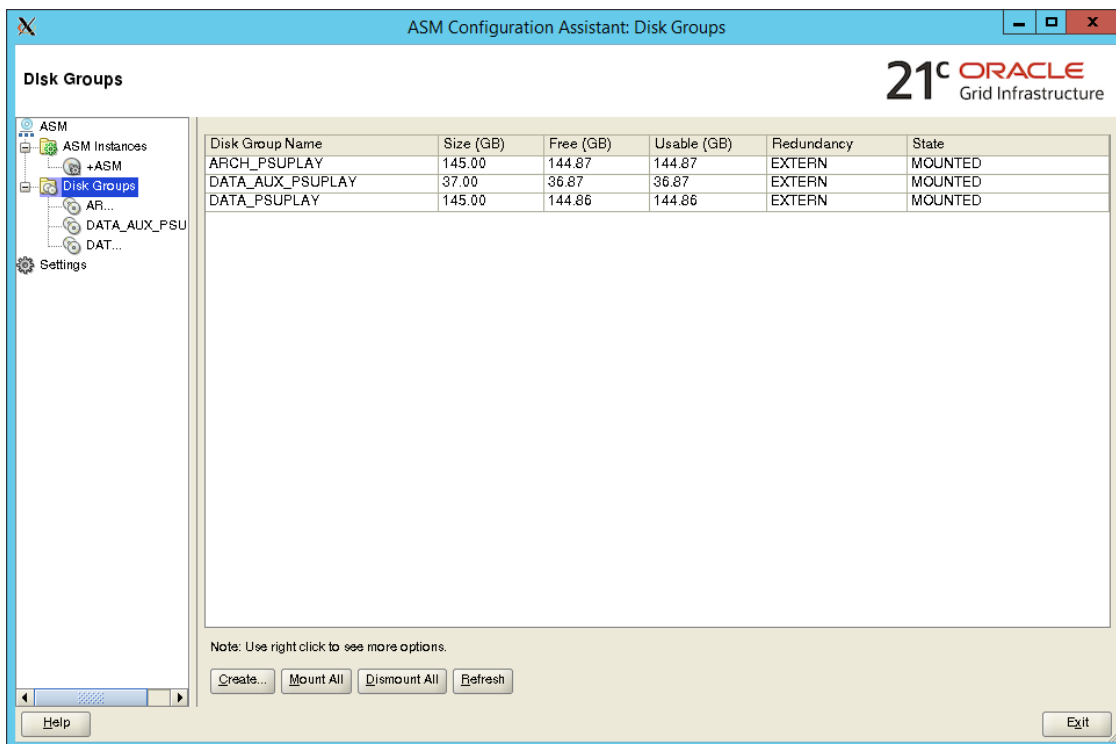
Enter the full pathname of the local bin directory: [/usr/local/bin]:
 The contents of "dbhome" have not changed. No need to overwrite.
 The contents of "oraenv" have not changed. No need to overwrite.
 The contents of "coraenv" have not changed. No need to overwrite.

Entries will be added to the /etc/oratab file as needed by
 Database Configuration Assistant when a database is created
 Finished running generic part of root script.
 Now product-specific root actions will be performed.

8- ASMCA, CRIAÇÃO DOS DISKGROUPS

Também, exatamente igual ao 19c

```
[grid@GNCASNHL08985 ~]$ . grid_profile
[grid@GNCASNHL08985 ~]$ export CV_ASSUME_DISTID=OEL8
[grid@GNCASNHL08985 ~]$ xauth add GNCASNHL08985/unix:10 MIT-MAGIC-COOKIE-1 554a7b51cb67a1c9ceee6635812cdd72
[grid@GNCASNHL08985 ~]$ export DISPLAY=localhost:10.0
[grid@GNCASNHL08985 ~]$ asmca
```

Validação no ASMCMD:

```
[grid@GNCASNHL08985 ~]$ asmcmd
ASMCMD> lsdg
State Type Rebal Sector Logical_Sector Block AU Total_MB Free_MB Req_mir_free_MB Usable_file_MB Offline_disks Voting_files
Name
MOUNTED EXTERN N 512 512 4096 4194304 148476 148344 0 148344 0 N ARCH_PSUPLAY
/
MOUNTED EXTERN N 512 512 4096 4194304 37884 37752 0 37752 0 N DATA_AUX_PSUP
LAY/
MOUNTED EXTERN N 512 512 4096 4194304 148476 148336 0 148336 0 N DATA_PSUPLAY/
ASMCMD> exit
[grid@GNCASNHL08985 ~]$
```

9 – DBCA, CRIAÇÃO DE UM BANCO EM PDB

Agora devemos ter a maior diferença quanto ao 19c, criar um banco em PDB:

```
[oracle@GNCASNHL08985 ~]$ . db_profile
[oracle@GNCASNHL08985 ~]$ export CV_ASSUME_DISTID=OL8
[oracle@GNCASNHL08985 ~]$ xauth add GNCASNHL08985/unix:10 MIT-MAGIC-COOKIE-1 554a7b51cb67a1c9ceee6635812cdd72
[oracle@GNCASNHL08985 ~]$ export DISPLAY=localhost:10.0
```

Database Configuration Assistant - Application - Step 1 of 14

Database Operation

Creation Mode

Deployment Type

Database Identification

Storage Option

Fast Recovery Option

Database Options

Configuration Options

Management Options

User Credentials

Creation Option

Summary

Progress Page

Finish

21c ORACLE Database

Select the operation that you want to perform.

Create a database

Configure an existing database

Delete database

Manage templates

Manage Pluggable databases

Oracle RAC database Instance management

Help

< Back

Next >

Finish

Cancel

Database Configuration Assistant - Create a database - Step 2 of 14

Database Operation

Creation Mode

Deployment Type

Database Identification

Storage Option

Fast Recovery Option

Database Options

Configuration Options

Management Options

User Credentials

Creation Option

Summary

Progress Page

Finish

21c ORACLE Database

Select Database Creation Mode

Typical configuration

Global database name: orcl

Storage type: Automatic Storage Management (ASM)

Database files location: +ARCH_P\$UPLAY{DB_UNIQUE_NAME} Browse...

Fast Recovery Area (FRA): +DATA_P\$UPLAY Browse...

Database character set: AL32UTF8 - Unicode UTF-8 Universal character set

Administrative password:

Confirm password:

Create as Container database

Pluggable database name:

Advanced configuration

Help

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Next >

Finish

Cancel

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Database Configuration Assistant - Create a database - Step 3 of 14

Database Operation

Creation Mode

Deployment Type

Database Identification

Storage Option

Fast Recovery Option

Database Options

Configuration Options

Management Options

User Credentials

Creation Option

Summary

Progress Page

Finish

Select the type of database you want to create.

Database type: Oracle Single Instance database

Database Management Policy: Automatic

Select a template for your database.

Templates that include datafiles contain pre-created databases. They allow you to create a new database quickly. Use templates without datafiles only when necessary, such as when you need to change attributes like block size that cannot be altered after database creation.

Template name	Include datafiles	Details
☒ General Purpose or Transaction Processing	Yes	View details
☐ Custom Database	No	View details
☐ Data Warehouse	Yes	View details

Template location: /u01/app/oracle/product/21.3/dbhome_1/assistants/dca/templates

Help

< Back

Next >

Finish

Cancel

Por padrão, coloquei o nome da instância, igual na 19c, neste caso, PSUPLAY, e criei a primeira pdb com sugestão da Oracle, pdb1:

Database Configuration Assistant - Create a database - Step 4 of 14

Database Operation

Creation Mode

Deployment Type

Database Identification

Storage Option

Fast Recovery Option

Database Options

Configuration Options

Management Options

User Credentials

Creation Option

Summary

Progress Page

Finish

Specify Database Identification Details

Provide a unique database identifier information. An Oracle database is uniquely identified by a Global database name, typically of the form "name.domain".

Global database name: PSUPLAY

SID: PSUPLAY

Service name:

☒ Create as Container database

A Container database can be used for consolidating multiple databases into a single database, and it enables database virtualization. A Container database (CDB) can have zero or more pluggable databases (PDB).

☒ Use Local Undo tablespaces for PDBs

☐ Create an empty Container database

☒ Create a Container database with one or more PDBs

Number of PDBs: 1

PDB name: pdb1

Help

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Next >

Finish

Cancel

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Database Configuration Assistant - Create 'PSUPLAY' database - Step 5 of 14

Database Operation

Creation Mode

Deployment Type

Database Identification

Storage Option

Fast Recovery Option

Database Options

Configuration Options

Management Options

User Credentials

Creation Option

Summary

Progress Page

Finish

☐ Use template file for database storage attributes

Storage type and location for database files will be picked up from the specified template (General Purpose or Transaction Processing).

☒ Use following for the database storage attributes

All the database files will be put at the specified location below. You can customize the name and location of each datafile in the subsequent screen.

Database files storage type: Automatic Storage Management (ASM)

Database files location: +DATA_PSUPLAY\{DB_UNIQUE_NAME}

Browse...

Oracle Managed files option will enable Oracle to automatically generate the names of the datafiles for simplified database management.

☒ Use Oracle-Managed Files (OMF)

Multiplex redo logs and control files...

File location variables...

Help

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Next >

Finish

Cancel

Database Operation

Creation Mode

Deployment Type

Database Identification

Storage Option

Fast Recovery Option

Database Options

Configuration Options

Management Options

User Credentials

Creation Option

Summary

Progress Page

Finish

Choose the recovery options for the database.

☐ Specify Fast Recovery Area

Recovery files storage type: Automatic Storage Management (ASM)

Fast Recovery Area: +DATA_PSUPLAY

Browse...

Fast Recovery Area size: 13896

MB

☐ Enable archiving

Edit archive mode parameters...

Help

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Next >

Finish

Cancel

21c

ORACLE

Database

21c

ORACLE

Database

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Database Configuration Assistant - Create 'PSUPLAY' database - Step 7 of 14

Specify Network Configuration Details

21c ORACLE Database

- Database Operation
- Creation Mode
- Deployment Type
- Database Identification
- Storage Option
- Fast Recovery Option
- Network Configuration**
- Configuration Options
- Management Options
- User Credentials
- Creation Option
- Summary
- Progress Page
- Finish

Listener selection

Listeners from Grid Infrastructure home and Database Oracle home are listed below. Specify the listener name and port to create a new listener in database Oracle home.

	Name	Port	Oracle home	Status
☒	LISTENER	1521	/u01/app/21.3/grid	Up

☐ Create a new listener

Listener name:

Listener port:

Oracle home:

Help

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Next >

Finish

Cancel

Database Configuration Assistant - Create 'PSUPLAY' database - Step 8 of 15

Select Oracle Data Vault Config Option

21c ORACLE Database

- Database Operation
- Creation Mode
- Deployment Type
- Database Identification
- Storage Option
- Fast Recovery Option
- Network Configuration
- Data Vault Option**
- Configuration Options
- Management Options
- User Credentials
- Creation Option
- Summary
- Progress Page
- Finish

☐ Configure Oracle Database Vault

Database Vault owner:

Password: Confirm password:

☐ Create a separate account manager

Account manager:

Password: Confirm password:

☐ Configure Oracle Label Security

☐ Configure Oracle Label Security with OJD

Help

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Next >

Finish

Cancel

Database Configuration Assistant - Create 'PSUPLAY' database - Step 9 of 15

Specify Configuration Options

Database Operation

Creation Mode

Deployment Type

Database Identification

Storage Option

Fast Recovery Option

Network Configuration

Data Vault Option

Configuration Options

Management Options

User Credentials

Creation Option

Summary

Progress Page

Finish

Memory

Sizing

Character sets

Connection mode

Use Automatic Shared Memory Management

SGA size:

4714

MB

PGA Size:

1572

MB

Use Manual Shared Memory Management

Shared pool size:

0

MB

Buffer cache size:

0

MB

Java pool size:

0

MB

Large pool size:

0

MB

PGA size:

0

MB

Total memory for database

0 MB

Use Automatic Memory Management

Memory target:

6286

MB

Help

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Next >

Finish

Cancel

Database Configuration Assistant - Create 'PSUPLAY' database - Step 9 of 15

Specify Configuration Options

Database Operation

Creation Mode

Deployment Type

Database Identification

Storage Option

Fast Recovery Option

Network Configuration

Data Vault Option

Configuration Options

Management Options

User Credentials

Creation Option

Summary

Progress Page

Finish

Memory

Sizing

Character sets

Connection mode

A block is the smallest unit of storage for allocation and for I/O. It cannot be changed once the database is created.

Block size:

8192

BYTES

Specify the maximum number of operating system user processes that can be simultaneously connected to this database. The value of this parameter includes the user processes and the Oracle background processes.

Processes:

320

Help

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Next >

Finish

Cancel

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Database Configuration Assistant - Create 'PSUPLAY' database - Step 9 of 15

Specify Configuration Options

21^c ORACLE Database

Database Operation

Creation Mode

Deployment Type

Database Identification

Storage Option

Fast Recovery Option

Network Configuration

Data Vault Option

Configuration Options

Management Options

User Credentials

Creation Option

Summary

Progress Page

Finish

MemorySizingCharacter setsConnection mode

☒ Dedicated server mode

The database will allocate a dedicated resource for each client connection in this mode. Use this mode when the number of total client connections is expected to be small or when clients will be making persistent, long-running requests to the database.

☐ Shared server mode

The database will use a shared pool of allocated resources for all client connections in this mode. Use this mode when a large number of users need to connect to the database simultaneously while efficiently utilizing system resources.

Specify the number of Shared Servers, which will be the number of processes that will be created when the instance is started.

Shared servers:

Help

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Next >

Finish

Cancel

Database Configuration Assistant - Create 'PSUPLAY' database - Step 10 of 15

Specify Management Options

Database Operation

Creation Mode

Deployment Type

Database Identification

Storage Option

Fast Recovery Option

Network Configuration

Data Vault Option

Configuration Options

Management Options

User Credentials

Creation Option

Summary

Progress Page

Finish

Specify the management options for the database.

☐ **Configure Enterprise Manager (EM) database express**

EM database express port:

5500

☐ Configure EM database-express port as global port

☐ **Register with Enterprise Manager (EM) cloud control**

QMS host:

QMS port:

EM admin username:

EM admin password:

ASMSNMP user password:

Help

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Next >

Finish

Cancel

Instalado com a senha: pgoldstandard2015

Database Configuration Assistant - Create 'PSUPLAY' database - Step 11 of 15

Specify Database User Credentials

Database Operation

Creation Mode

Deployment Type

Database Identification

Storage Option

Fast Recovery Option

Network Configuration

Data Vault Option

Configuration Options

Management Options

User Credentials

Creation Option

Summary

Progress Page

Finish

You must specify passwords for the following user accounts in the new database for security reasons.

☐ Use different administrative passwords

	Password	Confirm password
SYS		
SYSTEM		
PDBADMIN		

☒ Use the same administrative password for all accounts

Password:

Confirm password:

Help

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Next >

Finish

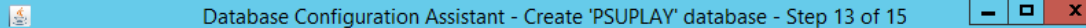
Cancel

Messages:

 Password: [DBT-06208] The 'ADMIN' password entered does not conform to the Oracle recommended standards.

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Summary

Database Operation

Creation Mode

Deployment Type

Database Identification

Storage Option

Fast Recovery Option

Network Configuration

Data Vault Option

Configuration Options

Management Options

User Credentials

Creation Option

Summary

Progress Page

Finish

21^c ORACLE Database

Database Configuration Assistant

Global Settings

Global database name: PSUPLAY

Configuration type: Oracle Single Instance database

SID: PSUPLAY

Create as Container database: Yes

Pluggable Database Name: pdb1

Number of Pluggable Databases: 1

Use Local Undo tablespace for PDBs: Yes

Database Files Storage Type: Automatic Storage Management (ASM)

Memory Configuration Type: Automatic Shared Memory Management

Template name: General Purpose

Initialization Parameters

audit_file_dest: {ORACLE_BASE}/admin/{DB_UNIQUE_NAME}/adump

audit_trail: db

compatible: 21.0.0

db_block_size: 8192 BYTES

db_create_file_dest: +DATA_F\$PSUPLAY/{DB_UNIQUE_NAME}/

db_name: PSUPLAY

diagnostic_dest: {ORACLE_BASE}

dispatchers: {PROTOCOL=TCP} (SERVICE=PSUPLAYXDB)

local_listener: LISTENER_P\$PSUPLAY

nls_language: AMERICAN

Save Response File...

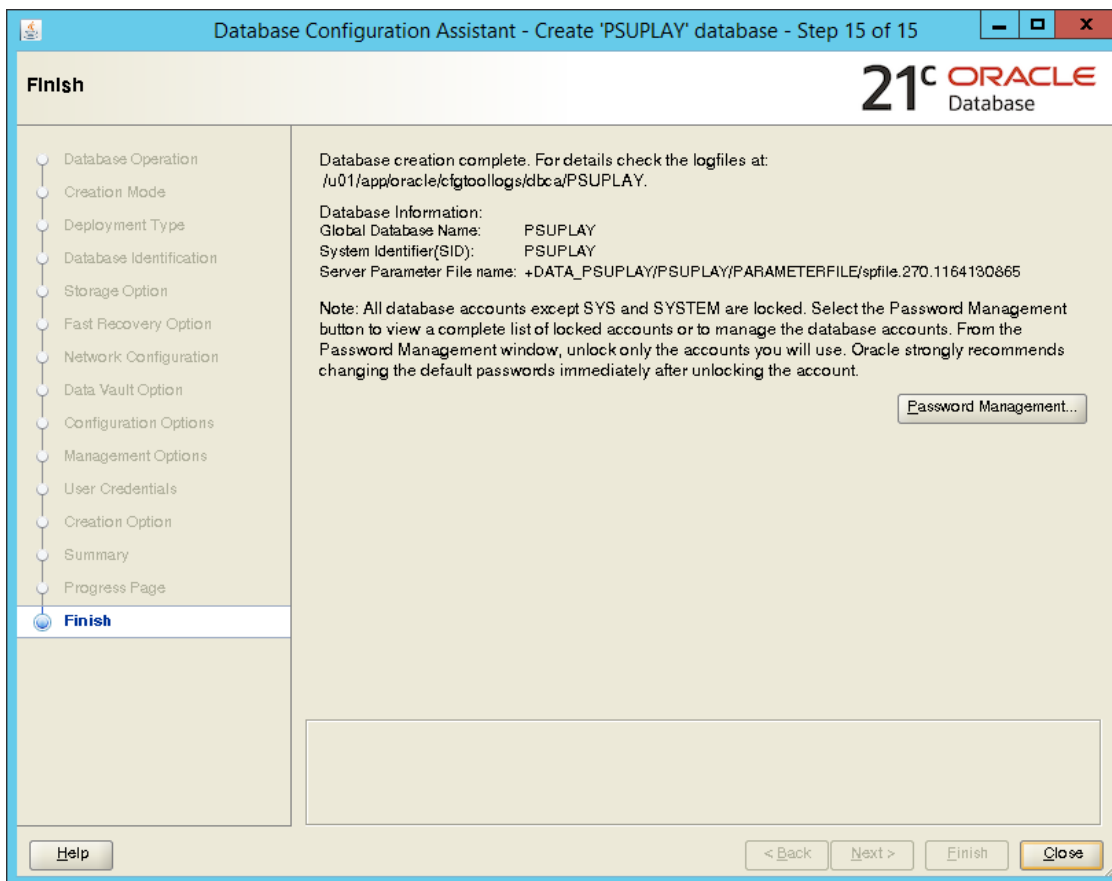
Help

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Next >

Finish

Cancel



Ajuste final:

```
[oracle@GNCASNHL08985 ~]$ sqlplus / as sysdba
```

```
SQL*Plus: Release 21.0.0.0.0 - Production on Wed Mar 20 17:45:32 2024
Version 21.3.0.0.0
```

```
Copyright (c) 1982, 2021, Oracle. All rights reserved.
```

```
Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0
```

```
SQL> show parameter db_create_file_dest
```

NAME	TYPE	VALUE
db_create_file_dest	string	+DATA_PSUPLAY

```
SQL> alter pluggable database pdb1 save state;
```

```
Pluggable database altered.
```

```
SQL> show con_name
```

```
CON_NAME
-----
CDB$ROOT
```

```
SQL> col name format a20
```

```
SQL> select con_id, name, open_mode from v$pdb;
```

CON_ID	NAME	OPEN_MODE
2	PDB\$SEED	READ ONLY
3	PDB1	READ WRITE

10 – BANCO EM ARCHIVELOG

```
SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> exit
Disconnected from Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

[oracle@GNCASNHL08985 ~]$ srvctl config database
PSUPLAY
[oracle@GNCASNHL08985 ~]$ srvctl config database -d PSUPLAY
Database unique name: PSUPLAY
Database name: PSUPLAY
Oracle home: /u01/app/oracle/product/21.3/dbhome_1
Oracle user: oracle
Spfile: +DATA_PSUPLAY/PSUPLAY/PARAMETERFILE/spfile.270.1164130865
Password file:
Domain:
Start options: open
Stop options: immediate
Database role: PRIMARY
Management policy: AUTOMATIC
Disk Groups: DATA_PSUPLAY
Services:
OSDBA group:
OSOPER group:
Database instance: PSUPLAY

[oracle@GNCASNHL08985 ~]$ srvctl start database -d PSUPLAY -o mount
[oracle@GNCASNHL08985 ~]$ sqlplus / as sysdba

SQL*Plus: Release 21.0.0.0.0 - Production on Wed Mar 20 17:53:33 2024
Version 21.3.0.0.0

Copyright (c) 1982, 2021, Oracle. All rights reserved.

Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL> alter system set log_archive_dest_1='LOCATION=+ARCH_PSUPLAY' scope=both sid='*';

System altered.

SQL> alter database archivelog;

Database altered.

SQL> archive log list
Database log mode                Archive Mode
Automatic archival               Enabled
Archive destination              +ARCH_PSUPLAY
Oldest online log sequence       1
Next log sequence to archive     3
Current log sequence             3
SQL> exit
Disconnected from Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0
[oracle@GNCASNHL08985 ~]$ srvctl stop database -d PSUPLAY
[oracle@GNCASNHL08985 ~]$ srvctl start database -d PSUPLAY
```